

Interview: Cassandra Extavour 467

22 Descent with Modification: A Darwinian View of Life 468

CONCEPT 22.1 The Darwinian revolution challenged traditional views of a young Earth inhabited by unchanging species 469

- Endless Forms Most Beautiful 469
- Scala Naturae* and Classification of Species 470
- Ideas About Change over Time 470
- Lamarck's Hypothesis of Evolution 471

CONCEPT 22.2 Descent with modification by natural selection explains the adaptations of organisms and the unity and diversity of life 471

- Darwin's Research 471
- Ideas from *The Origin of Species* 473
- Key Features of Natural Selection 476

CONCEPT 22.3 Evolution is supported by an overwhelming amount of scientific evidence 476

- Direct Observations of Evolutionary Change 477
- Homology 479
- The Fossil Record 481
- Biogeography 482
- What Is Theoretical About Darwin's View of Life? 483



23 The Evolution of Populations 486

CONCEPT 23.1 Genetic variation makes evolution possible 487

- Genetic Variation 487
- Sources of Genetic Variation 488

CONCEPT 23.2 The Hardy-Weinberg equation can be used to test whether a population is evolving 489

- Gene Pools and Allele Frequencies 490
- The Hardy-Weinberg Equation 490

CONCEPT 23.3 Natural selection, genetic drift, and gene flow can alter allele frequencies in a population 493

- Natural Selection 494
- Genetic Drift 494
- Gene Flow 496

CONCEPT 23.4 Natural selection is the only mechanism that consistently causes adaptive evolution 497

- Natural Selection: *A Closer Look* 497
- The Key Role of Natural Selection in Adaptive Evolution 498

- Sexual Selection 499
- Balancing Selection 500
- Why Natural Selection Cannot Fashion Perfect Organisms 501

24 The Origin of Species 506

CONCEPT 24.1 The biological species concept emphasizes reproductive isolation 507

- The Biological Species Concept 507
- Other Definitions of Species 510

CONCEPT 24.2 Speciation can take place with or without geographic separation 511

- Allopatric ("Other Country") Speciation 511
- Sympatric ("Same Country") Speciation 513
- Allopatric and Sympatric Speciation: *A Review* 516

CONCEPT 24.3 Hybrid zones reveal factors that cause reproductive isolation 516

- Patterns Within Hybrid Zones 516
- Hybrid Zones and Environmental Change 517
- Hybrid Zones over Time 518

CONCEPT 24.4 Speciation can occur rapidly or slowly and can result from changes in few or many genes 520

- The Time Course of Speciation 520
- Studying the Genetics of Speciation 522
- From Speciation to Macroevolution 523

25 The History of Life on Earth 525

CONCEPT 25.1 Conditions on early Earth made the origin of life possible 526

- Synthesis of Organic Compounds on Early Earth 526
- Abiotic Synthesis of Macromolecules 527
- Protocells 527
- Self-Replicating RNA 528

CONCEPT 25.2 The fossil record documents the history of life 528

- The Fossil Record 529
- How Rocks and Fossils Are Dated 529
- The Origin of New Groups of Organisms 530

CONCEPT 25.3 Key events in life's history include the origins of unicellular and multicellular organisms and the colonization of land 532

- The First Single-Celled Organisms 533
- The Origin of Multicellularity 535
- The Colonization of Land 536

CONCEPT 25.4 The rise and fall of groups of organisms reflect differences in speciation and extinction rates 537

- Plate Tectonics 538
- Mass Extinctions 540
- Adaptive Radiations 542

CONCEPT 25.5 Major changes in body form can result from changes in the sequences and regulation of developmental genes 544

- Effects of Developmental Genes 544
- The Evolution of Development 545

CONCEPT 25.6 Evolution is not goal oriented 547

- Evolutionary Novelties 547
- Evolutionary Trends 548

